Om Prakash Singh

Last updated in March 2026

Hyderabad, India | omprakash300@gmail.com | 7415741371 | omsingh | omsingh-nitd | fantops

Professional Summary

Silicon Validation Engineer with 8 years of pre-silicon and post-silicon validation experience across Intel and Microsoft, specializing in PCIe, power management, and SoC-level debugging. Currently validating AI accelerator SoCs at Microsoft, with hands-on understanding of on-device model execution pipelines (ONNX Runtime, inference latency). Uses agentic AI workflows (GitHub Copilot CLI) to accelerate trace analysis and debug triage. Proven at defining validation strategy, debugging complex cross-IP issues, and C/C++ test development on FPGA-based emulation (Zebu) and JTAG.

Experience

Device Validation Engineer 2 (Post-Silicon Validation) *Hyderabad, India*
Microsoft *June 2024 – Present*

- Owned post-silicon validation of AI accelerator SoCs (Intel, AMD, Qualcomm) — defined test strategy for NPU/CPU integration, power state transitions, and feature correctness on production silicon
- Built EdgeOptimizer, a post-silicon instrumentation tool to characterize CPU/NPU utilization and inference latency on real silicon, surfacing resource bottlenecks in AI accelerator workloads
- Drove root cause analysis using WPR/WPA silicon-level traces and ETL logs — accelerated triage by deploying agentic AI workflows (GitHub Copilot CLI) to parse and correlate trace data, reducing debug cycle time for concurrency failures, scheduling anomalies, and silicon bugs across CPU/NPU execution paths

Emulation Engineer (Pre-Silicon Validation) *Bangalore, India*
Intel *July 2021 – June 2024*

- Owned pre-silicon validation of power management flows (C-states, ADR) and PCIe Gen6 on Xeon server SoCs — defined coverage plans from CPU architecture specs and drove silicon readiness milestones
- Developed 10+ PCIe Gen6 transactor enhancements in C++ and validation content using Perspec on Zebu (FPGA) emulation platforms; debugged 15+ cross-IP issues using bare-metal C
- Represented validation in cross-functional architecture reviews; contributed to ˜25% reduction in post-silicon escapes

Software Engineer (Post-Silicon Validation) *Bangalore, India*
L&T Technology Services *Sep 2018 – July 2021*

- Owned post-silicon bring-up and validation of NVMe SoC platforms from first power-on through production readiness, using JTAG (Metaware) and Lecroy protocol analyzers
- Built Python automation suite covering 25+ power management test scenarios; diagnosed 10+ power-related silicon bugs through cross-functional firmware debug
- Validated PCIe/NVMe protocol compliance and power state transitions; led SDExpress POC validation effort

Education

- **M.Tech in Computer Science & Engineering**, National Institute of Technology, Durgapur – 2018
- **B.Tech in Computer Science & Engineering**, RGPV, Bhopal – 2015

Technical Skills

- **Programming & Scripting:** C, C++, Python, Linux, PowerShell
- **Validation & Debug:** Pre-Silicon Validation, Post-Silicon Validation, JTAG, Protocol Debug (PCIe, CXL, NVMe)
- **Systems & Architecture:** SoC Architecture, CPU Subsystem, PCIe, Power Management, AI Accelerator
- **Tools:** WPR/WPA, WinDbg, Oscilloscope, Lecroy Protocol Analyzer, Linux Debug Tools
- **AI & Agentic Workflows:** Agentic AI Workflows, GitHub Copilot CLI, On-Device Models(ONNX Runtime)
- **Domains:** Silicon Validation, SoC Integration, Power Management Validation, PCIe Validation

Awards & Recognition

- Intel: Star of the Month for CXL firmware validation and early bug discovery
- L&T/SanDisk: Spot Awards for SDExpress POC validation